

The Rocket Flight Database: An Externally Curated Supersonic Sounding-Rocket and High-Power-Rocketry Apogee Corpus with Paired Commercial-Simulator Predictions

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Abstract

The Rocket Flight Database is an externally curated corpus of instrumented sounding-rocket and high-power-rocketry flights assembled to benchmark supersonic trajectory simulators. The headline table comprises 25 flights spanning Mach 0.54–4.33, each pairing a measured apogee (barometric, GPS, optical-track, or integrated-accelerometer ground truth) with two predictions: the publicly recorded RASAero II [1] altitude and an OpenRocket-Plus [2] re-simulation. A second table reports approximately 20 exploratory high-Mach historical sounding-rocket flights (to Mach 7.2) reproduced in full, without outcome-based selection. Each record carries per-flight provenance to the originating source document. The data are released CC-BY-4.0 at Zenodo (DOI 10.5281/zenodo.19976138). Their reuse value is as an open, non-circular benchmark for validating rocketry trajectory and aerodynamics codes [3]; the underlying accuracy analysis is reported in the companion research article [4].

Keywords: rocket flight data, apogee, supersonic aerodynamics, trajectory simulation, OpenRocket, RASAero II, model validation

Table 1: Specifications Table.

Subject	Aerospace Engineering
Specific subject area	Supersonic rocket aerodynamics and trajectory simulation; apogee prediction and model validation for sounding rockets and high-power amateur rocketry.
Type of data	Table (comma-separated values, CSV); plain-text provenance documentation (Markdown). The reusable observational core is the externally measured flight apogees (raw) together with the independently published RASAero II reference predictions; the this-work apogee columns are reproducible simulated predictions supplied as paired benchmark metadata (processed), not the primary observed datum.

How the data were acquired	Externally measured flight ground truth was compiled from published instrumented flight records: barometric altimeter, GPS, optical (theodolite) track, integrated accelerometer, and radar/beacon tracking. Paired commercial-simulator reference apogees are the recorded RASAero II [1] altitude-comparison predictions published by Rogers Aeroscience [5]. The OpenRocket-Plus (<code>apogee_thiswork_ft</code>) column was generated by running the archived simulator release on each reconstructed vehicle; predicted apogees are read directly from the simulation output. The reusable observational core of the corpus is therefore the externally measured flight apogees plus the independently published RASAero II predictions (both observation/external); the this-work apogee column is a reproducible simulated prediction supplied as paired benchmark metadata, not the primary observed datum.
Data format	Raw and analysed. Two CSV tables (<code>flight_comparison.csv</code> , <code>sounding_rockets_exploratory.csv</code>) and one Markdown provenance/reproducibility statement (<code>reproducibility.md</code>).

Description of data collection	The headline corpus comprises 25 externally curated flights spanning Mach 0.54–4.33 (23 single-stage high-power flights plus two two-stage vehicles, the AeroPac 104K Two-Stage at Mach 3.04 and the MESOS 293K closure at Mach 4.33), each pairing a measured apogee with the corresponding RASAero II reference prediction. A separate exploratory set of ~ 20 historical high-Mach sounding-rocket flights (e.g., Nike-Apache, Nike-Deacon, Nike-Cajun, Arcas, Black Brant V, HEROS 3) is reported in full, with primary flight-report citations per row. The headline corpus was selected externally (it is Rogers’ public RASAero II comparison set), not curated by outcome.
Data source location	Flights are drawn from published sources documented per record in the dataset. Headline measured and RASAero II values: Rogers Aeroscience RASAero II altitude-comparison set [5]. Exploratory ground truth: original NACA/NASA/DTIC flight reports (cited per row, e.g., [6, 7, 8]). Geographic launch sites are recorded per flight (e.g., Black Rock Desert, NV, USA; Wallops Island, VA, USA; White Sands, NM, USA; Esrange, Sweden; Churchill, MB, Canada).
Data accessibility	Repository name: Rocket Flight Database. Data identification number (DOI): 10.5281/zenodo.19976138 (version of record at submission). Direct URL to data: https://doi.org/10.5281/zenodo.19976138. Source-code/development mirror: https://github.com/AidanSYu/rocket-flight-database. License: Creative Commons Attribution 4.0 International (CC-BY-4.0) [9].

Related research
article

A. Yu, “Integrated open-source supersonic aerodynamics for sounding-rocket and high-power apogee prediction: a 25-flight validation and architecture,” *Journal of Spacecraft and Rockets* (companion research article, under review) [4]. The present dataset is the validation corpus described and analysed in that article; this Data in Brief article describes the data only and draws no new scientific conclusions.

Value of the Data

- The database is, to the authors’ knowledge, the first openly published, externally curated apogee benchmark for supersonic sounding rockets and high-power rocketry: the 25-flight headline corpus is drawn from a third-party RASAero II altitude-comparison collection [1], so flight selection is independent of any outcome of the simulator described in the parent article [4].
- Each flight is supplied with a paired commercial-baseline prediction (RASAero II [1]) alongside the measured apogee, enabling a head-to-head, apples-to-apples evaluation of any rocket-trajectory simulator against both ground truth and an established reference tool without re-deriving the baseline.
- Every reported statistic is reproducible: the published analysis scripts regenerate all signed errors, summary statistics, and bootstrap confidence intervals directly from the full-precision apogee columns, so reusers can audit and re-derive every figure rather than trusting pre-computed values.
- The corpus spans the subsonic, transonic, and supersonic regimes from Mach 0.54 to Mach 4.33, with an exploratory extension of roughly 20 historical high-Mach sounding-rocket flights reaching toward Mach 7 (e.g. Black Brant V VB at Mach 7.22), giving a continuous Mach ladder for regime-by-regime assessment.
- The data are reusable for validating or benchmarking any rocket aerodynamics or trajectory tool—including open-source codes such as Open-

Rocket [2], RocketPy [3], and engineering-level prediction methods [10]—against a common, version-locked set of flights.

- Each record carries vehicle, motor, body-diameter, launch-site altitude, and instrumentation-type metadata, together with the cited source document, so reusers can reconstruct and re-simulate the flights or trace every value back to its original published record.

1. Objective

Validating a flight-trajectory simulator in the supersonic regime requires a ground-truth corpus that pairs measured flight outcomes with the predictions of an established reference tool. For hobby- and sounding-rocket aerodynamics, no such corpus has been openly available: published flight records are scattered across primary sources in differing formats, and the most widely used commercial predictions for these vehicles—those produced by RASAero II [1]—are not distributed as a machine-readable, reusable table alongside the corresponding measured apogees. As a result, comparisons between open-source simulators such as OpenRocket [2] and a commercial baseline have typically been made on small, internally selected sets of flights, which makes the resulting accuracy statistics difficult to reproduce and easy to bias through outcome-based selection.

The Rocket Flight Database addresses this gap. Its headline component is a 25-flight corpus spanning Mach 0.54–4.33, drawn from Rogers’ publicly published RASAero II altitude-comparison collection (23 single-stage high-power flights plus two two-stage flights—the AeroPac 104K Two-Stage at Mach 3.04 and the MESOS 293K flight, the highest-Mach case at Mach 4.33). Because the flights were selected externally—by the authors of the commercial comparison set rather than curated by us for favourable outcomes—and because each record carries both the measured apogee and the paired RASAero II prediction, the table supports an outcome-independent, reproducible validation of supersonic apogee prediction. The ground-truth column reports instrumented apogees (barometric, GPS, optical-track, and integrated-accelerometer records), and the commercial-reference column reproduces Rogers’ recorded RASAero II predictions verbatim; only the OpenRocket-Plus prediction column is regenerated from the archived simulator code, so the comparison is non-circular.

A companion exploratory table extends the database to roughly twenty historical high-Mach sounding-rocket flights (Black Brant V, Nike–Apache,

Nike–Cajun, Nike–Deacon, Arcas, and HEROS 3), reported in full—including the cases that fall well outside the headline admission band—to document the simulator’s behaviour beyond Mach 4.33 without selection bias.

This dataset is the validation corpus underlying the companion research article that introduces OpenRocket-Plus, an open-source supersonic extension of OpenRocket [2], and reports its integrated apogee accuracy against the RASAero II baseline [4]. That parent article presents and interprets the comparison result; the present article is descriptive and confines itself to the data’s schema, provenance, and reuse value. Releasing the corpus openly under a permissive licence lets other investigators independently reproduce those statistics, benchmark alternative tools—including newer open-source trajectory frameworks such as RocketPy [3] and recent amateur-supersonic flight studies [11, 12]—and extend the corpus with additional instrumented flights.

2. Data Description

The Rocket Flight Database is distributed as two comma-separated-value (CSV) files together with a plain-text reproducibility statement. The first file, `flight_comparison.csv`, is the headline corpus: 25 instrumented flights with paired apogee predictions from a commercial trajectory simulator (RASAero II [1, 5]) and from the open-source predictor described in the companion research article [4]. The second file, `sounding_rockets_exploratory.csv`, is a separate exploratory set of historical high-Mach sounding-rocket flights that is deliberately held outside the headline corpus. Throughout both files, all apogee values are reported in feet (ft) and all error values in percent (%); peak Mach is dimensionless and body diameter is in inches (in). No values in the two distributed files are rounded beyond the precision shown; the single-decimal `err*_pct` columns are convenience columns recomputed downstream from the full-precision apogee columns (see `reproducibility.md`).

2.1. File 1: *flight_comparison.csv* (25-flight headline corpus)

Each row is one flight. The 14 columns are defined in Table 2. The corpus spans peak Mach 0.54 to 4.33 and body diameters from 1.75 in to 8.00 in; two flights are two-stage vehicles for which a slash-separated pair of stage diameters and motor designations is recorded (AeroPac 104K Two-Stage and MESOS 293K).

Table 2: Column dictionary for `flight_comparison.csv`.

Column	Type	Unit	Description
<code>flight_id</code>	integer	—	Sequential record identifier (1–25).
<code>vehicle_name</code>	string	—	Rocket / project name as published in the source comparison set.
<code>motor</code>	string	—	Motor designation(s) and manufacturer; a slash separates per-stage motors for two-stage vehicles.
<code>diameter_in</code>	float	in	Body reference diameter; a slash separates per-stage diameters for two-stage vehicles.
<code>peak_mach</code>	float	—	Peak flight Mach number.
<code>launch_site_alt_ft</code>	integer	ft	Launch-site elevation above mean sea level.
<code>apogee_real_ft</code>	integer	ft	Measured (ground-truth) apogee.
<code>apogee_rasaero_ft</code>	integer	ft	Apogee predicted by RASAero II (commercial reference).
<code>apogee_thiswork_ft</code>	integer	ft	Apogee predicted by the open-source predictor of this work.
<code>err_rasaero_pct</code>	float	%	Signed relative apogee error of RASAero II vs. measured.
<code>err_thiswork_pct</code>	float	%	Signed relative apogee error of this work vs. measured.
<code>abs_err_delta_pp</code>	float	pp	$ \text{err_rasaero} - \text{err_thiswork} $ in percentage points (paired absolute-error difference).
<code>flight_data_type</code>	string	—	Instrumentation class of the measured apogee (see below).
<code>data_source</code>	string	—	Origin of the record (the published RASAero II altitude-comparison set).

The `flight_data_type` column records how the measured apogee was obtained, taking one of four values in this file:

- **Barometric Altimeter** — apogee from an onboard pressure altimeter (17 of 25 flights);
- **GPS** — apogee from an onboard GPS receiver (3 flights: A-601 Kinsel, AeroPac 104K Two-Stage, MESOS 293K);
- **Optical Track** — apogee from ground-based optical tracking (3 flights: Kline-Rogers L500 and the two Full Metal Jacket flights);
- **Integrated Accelerometer** — apogee from twice-integrated onboard accelerometer data (2 flights: Proteus 6 and Qu8k).

The measured-truth, RASAero II, and “this work” columns are produced independently of one another, so the per-flight comparison is non-circular; the provenance split is documented in `reproducibility.md`.

Table 3 reproduces the 25-flight corpus verbatim from `flight_comparison.csv`: vehicle, peak Mach, measured apogee, and the RASAero II and this-work apogees with their signed percent errors. All apogees are in feet and all errors in percent.

Table 3: The 25-flight headline corpus (`flight_comparison.csv`), reproduced verbatim. Apogees in feet; errors in percent (signed, vs. measured).

ID	Vehicle	Peak Mach	Measured apogee (ft)	RAS apogee (ft)	RAS err (%)	This work apogee (ft)	This work err (%)
1	Thunder & Lightning	0.54	3577	3989	11.52	3877	8.4
2	Gibb	0.55	3913	4310	10.15	3989	1.9
3	Cancer Descending	0.56	6188	6328	2.26	6044	−2.3
4	EZI-65 J450ST	0.60	3965	4214	6.28	4158	4.9
5	Caliber Isp 04 AVTC Team 2	0.64	3710	3871	4.34	3890	4.9
6	Caliber Isp 04 AVTC Team 3	0.64	3964	3871	−2.35	3889	−1.9
7	Caliber Isp 04 AVTC Team 1	0.66	3837	3943	2.76	3960	3.2
8	Byrum	0.75	5732	5280	−7.89	6161	7.5

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Table 3 (continued)

ID	Vehicle	Peak Mach	Measured apogee (ft)	RAS apogee (ft)	RAS err (%)	This work apogee (ft)	This work err (%)
9	Ion Drive	0.79	8027	8642	7.66	7730	−3.7
10	Caliber Isp 05 ARO-414 (Discovery)	0.81	4930	4831	−2.01	4772	−3.2
11	Blister	0.83	9026	8347	−7.52	8268	−8.4
12	Caliber Isp 05 ARO-414 (Columbia)	0.84	5085	4842	−4.78	4777	−6.1
13	Rabia - Short Fin Can	0.86	10584	10376	−1.97	9916	−6.3
14	Raven	1.07	8815	9288	5.37	9489	7.6
15	Rabia	1.14	12745	12777	0.25	11913	−6.5
16	Torrent	1.22	12807	13852	8.16	12455	−2.8
17	Kline-Rogers L500	1.98	24771	26485	6.92	24179	−2.4
18	A-601 Kinsel	2.19	42771	41086	−3.94	46499	8.7
19	Full Metal Jacket - BALLS 005	2.31	37981	38820	2.21	37256	−1.9
20	Full Metal Jacket - Black Rock 6	2.46	30038	32646	8.68	29239	−2.7
21	Proteus 6	2.87	85067	86799	2.04	91339	7.4
22	AeroPac 104K Two-Stage	3.04	104659	113786	8.72	103602	−1.0
23	Don't Debate This	3.04	56573	62308	10.14	53150	−6.1
24	Qu8k	3.46	121478	116254	−4.30	119187	−1.9
25	MESOS 293K	4.33	293488	289789	−1.26	273056	−6.96

All 25 flights have a measured apogee, a RASAero II apogee, and a this-work apogee. The corpus contains no flight above Mach 4.33; the highest-Mach record is the two-stage MESOS 293K closure (peak Mach 4.33, measured apogee 293,488 ft), for which this work records an apogee of 273,056 ft (−6.96%). The value-of-the-data and detailed comparison statistics are presented in the companion research article [4]; this article describes the data only.

2.2. File 2: *sounding_rockets_exploratory.csv* (exploratory high-Mach set)

The second file holds 20 historical high-Mach sounding-rocket flights simulated with the same open-source predictor and reported in full as an ex-

ploratory capability set that is excluded from the headline corpus. Each row is one flight; the eight columns are defined in Table 4.

Table 4: Column dictionary for `sounding_rockets_exploratory.csv`.

Column	Type	Unit	Description
<code>vehicle</code>	string	—	Vehicle / flight designation.
<code>source_doc</code>	string	—	Primary published source for the measured apogee (radar / beacon / GPS), cited per row.
<code>real_apogee_ft</code>	integer	ft	Measured (ground-truth) apogee.
<code>orp_apogee_ft</code>	integer	ft	Apogee predicted by the open-source predictor of this work (blank for transparently reported simulation-error rows).
<code>err_thiswork_pct</code>	float	%	Signed relative apogee error of this work vs. measured (blank where no apogee was produced).
<code>peak_mach</code>	float	—	Peak flight Mach number (blank where no trajectory was produced).
<code>headline_admissible</code>	string	—	Whether the flight falls within $\pm 10\%$; values are “within $\pm 10\%$ ”, “NO ($> 10\%$ error)”, or “NO (sim error)”.
<code>note</code>	string	—	Free-text note recording solver-termination and reconstruction caveats (e.g. ascent-only cutoff, descent NaN, zero-apogee, simulation error).

Of the 20 exploratory flights, 3 fall within $\pm 10\%$ of the measured apogee: Black Brant V VB (AAF-VB-32) at peak Mach 7.224 with error -6.97% [8], Nike-Deacon no. 1 at peak Mach 4.956 with error -1.06% , and Nike-Deacon no. 2 at peak Mach 5.079 with error -0.89% . The remaining 17 flights fall outside $\pm 10\%$ and are reported transparently: the Nike-Apache family over-predicts ($+24$ to $+36\%$), the Nike-Cajun University of Michigan sounding flight over-predicts ($+16.56\%$), and the Arcas (blunt and secant-ogive) and HEROS 3 flights under-predict (-29 to -69%). This count of 17 also includes two transparently reported non-nominal solver outcomes: one zero-apogee record (Nike-Apache NASA 14.210 GI, `orp_apogee_ft` = 0, `err_thiswork_pct` = -100.0 , flagged “ZERO APOGEE” in the note) and one simulation-error record (Nike-Cajun Hurricane, with `orp_apogee_ft`,

`err_thiswork_pct` and `peak_mach` left blank and marked “NO (sim error)”. As with File 1, all apogees are in feet and all errors in percent. The full set is published here so that the exploratory outcomes are reported without selection.

3. Experimental Design, Materials and Methods

This section documents how each column of the database was produced. No new analysis or scientific interpretation is presented here; those belong to the companion research article [4], which this data article accompanies. The design principle is that the three quantity groups in `flight_comparison.csv` — measured ground truth, the commercial reference prediction, and the open-source prediction of this work — originate from three independent sources, so that the paired comparison is non-circular. The measured and commercial-reference columns are never regenerated locally; only the `apogee_thiswork_ft` column is produced by the archived code release described below.

3.1. *Measured ground truth (real apogees and flight metadata)*

The columns `apogee_real_ft`, `peak_mach`, `flight_data_type`, `data_source`, `diameter_in`, and `launch_site_alt_ft` are externally measured quantities transcribed from published flight records; they are not computed by this work. For the 23 single-stage flights of the headline corpus, the apogees and the associated flight metadata are the values published in C. E. Rogers’ RASAero II altitude-comparison collection [1, 5], which records each flight together with the instrumentation used to determine apogee (barometric altimeter, optical track, GPS, or integrated accelerometer; see the `flight_data_type` column). The two two-stage vehicles—the AeroPac 104K Two-Stage and MESOS 293K—take their measured apogees (104,659 ft and 293,488 ft, respectively) from the flight record in the same way. The heterogeneity of the measurement methods across the corpus implies a non-uniform ground-truth uncertainty floor of order one percent, which is reported as-is and not subtracted from any error figure.

For the exploratory high-Mach set in `sounding_rockets_exploratory.csv`, the real apogees in `real_apogee_ft` are radar-, beacon-, or GPS-tracked values taken from the primary historical document cited per row in the `source_doc` field (e.g. the NACA Nike–Deacon flight report [6] and the Black Brant VB flight report [8]; the Nike–Apache radar apogees are taken from the NASA Goddard Wallops radar-apogee report X-721-67-103 named

in those rows’ `source_doc` field). Each row’s provenance is therefore self-contained in `source_doc`; the corresponding provenance statement is given in the repository’s `reproducibility.md`.

3.2. Commercial reference predictions (*RASAero II*)

The columns `apogee_rasaero_ft` and `err_rasaero_pct` are the RASAero II [1] altitude predictions *as recorded by C. E. Rogers* in his public RASAero II altitude-comparison set [5]. They are transcribed from that published source and were *not* re-run by this work. Consequently they are version-locked recorded predictions rather than fresh, independently re-run pre-flight cases; the companion research article [4] frames the comparison accordingly. Reproducing these specific predictions would require the original RASAero II configuration files and software version used by Rogers, which are outside the scope of this dataset.

3.3. This-work predictions (*OpenRocket-Plus*)

The columns `apogee_thiswork_ft`, `err_thiswork_pct`, and `abs_err_delta_pp` are produced by OpenRocket-Plus, the supersonic extension of the open-source OpenRocket simulator [2], as embodied in its archived code release. Each flight is represented by a per-vehicle `.ork` model built from the geometry, mass, and motor specification given in the vehicle’s source documentation; for the two-stage MESOS 293K closure the model uses reconstructed KIP-EX motors. The `err_thiswork_pct` value is the signed relative error of the predicted apogee against the measured apogee in the same row, and `abs_err_delta_pp` is the paired difference of absolute errors, RASAero II minus this work, $|\text{err_rasaero_pct}| - |\text{err_thiswork_pct}|$, in percentage points.

At a high level, each `.ork` model is flown through the standard OpenRocket six-degree-of-freedom flight-dynamics integrator with the launch-site elevation set from `launch_site_alt_ft` and a standard atmosphere [13]. The aerodynamic build extends the Barrowman-type method [14] of baseline OpenRocket [2] with the supersonic and base-drag models documented in the companion research article [4]; the engineering provenance of those component models is recorded there and in the related software and technical-report companions. Predicted apogee is read from the simulation output. The reported `peak_mach` for each flight characterizes the regime reached.

3.4. The MESOS 293K model value

For the highest-Mach flight in the corpus, the two-stage MESOS 293K closure, the present archived code release predicts 273,056 ft, an error of -6.96% against the measured 293,488 ft; this is the value carried in the published CSV and used throughout the companion article [4]. It is the corpus’s largest single-flight error and is reproducible: running the MESOS flight in isolation under the present archived release reproduces 273,056 ft. An earlier deposited version of this record carried a different value for this single flight; that value was erroneous, has no defensible derivation, and is superseded by the version of record, whose model column matches the present archived code. The other 24 model values are unchanged. Because -6.96% remains within the $\pm 10\%$ admission band, the 25/25-within- $\pm 10\%$ membership of the headline corpus is unaffected.

3.5. Reproducibility of the derived statistics

The two `err*_pct` columns and `abs_err_delta_pp` are stored rounded to limited precision; all reported summary statistics are recomputed from the full-precision `apogee*_ft` columns, so the apogee values are the single source of truth. Two Python scripts distributed with the dataset regenerate every derived figure from the published `flight_comparison.csv`: `analyze.py` recomputes the corpus bias/variance summary (mean signed error, standard deviation, RMSE, MAE, and the $\pm 5\%/\pm 10\%$ counts), and `uncertainty_quantification.py` regenerates the bootstrap confidence intervals and the paired RASAero II comparison (including the Wilcoxon signed-rank test on absolute error [15]). The reproduction recipe, software versions, and seed are listed in the repository’s `reproducibility.md`. The interpretation of these statistics — the unbiasedness of the mean error, the statistical-parity finding versus RASAero II, and the per-regime and in-sample disclosures — is presented and defended in the companion research article [4] and is deliberately not repeated here.

4. Limitations

The following limitations bound the reuse of this dataset. They are stated descriptively; the accuracy result they qualify is interpreted in the companion research article [4], which this data article accompanies and which should be consulted for any scientific conclusion drawn from the tables.

Heterogeneous ground truth and an unsubtracted measurement floor.. The measured apogees in `flight_comparison.csv` are not produced by a single instrument or campaign: the `flight_data_type` column records barometric-altimeter, GPS, optical-tracking, and integrated-accelerometer determinations across high-power-rocketry flights of differing instrumentation and vintage, while the exploratory sounding-rocket apogees in `sounding_rockets_exploratory.csv` are radar-, beacon-, and GPS-tracked historical records. Each technique carries its own bias and dispersion, so the ground-truth column embeds an irreducible measurement uncertainty of order 1%. This floor is reported but deliberately *not* subtracted from the model errors: the published `err_thiswork_pct` and `err_rasaero_pct` values are therefore conservative, in the sense that some of the residual they attribute to the predictors is in fact ground-truth noise. Users comparing any predictor against this corpus should treat sub-percent differences as within the noise of the reference data.

Reconstruction uncertainty dominates the exploratory high-Mach set.. The headline corpus (`flight_comparison.csv`, 25 flights, Mach 0.54–4.33) contains no flight above Mach 4.33, and the high-Mach behavior of the method is documented only by the separate, explicitly exploratory `sounding_rockets_exploratory.csv` table. Of the ~ 20 historical sounding-rocket flights in that file, only three fall within $\pm 10\%$ of the recorded apogee — Black Brant V VB (Mach 7.224, -6.97%), Nike-Deacon no. 1 (Mach 4.956, -1.06%), and Nike-Deacon no. 2 (Mach 5.079, -0.89%). The remainder scatter widely: the Nike-Apache family over-predicts systematically ($+24$ to $+36\%$) and the Arcas and HEROS 3 vehicles under-predict strongly (-29 to -69%), with a small number of simulation-error or zero-apogee cases reported transparently rather than removed. This spread is driven far more by motor-thrust and geometry *reconstruction* uncertainty in the poorly documented historical source material than by the aerodynamic model: these flights are provided so that the full, uncurated picture is on record, and they must *not* be read as a flight-validation result. Reporting only the three admissible cases would constitute outcome-based selection; the entire set is published for this reason.

RASAero II values are recorded predictions, not independent reruns.. The `apogee_rasaero_ft` and `err_rasaero_pct` columns are the predictions *as published* in Rogers’ public RASAero II altitude-comparison collection [1, 5]; they are reproduced verbatim and were *not* regenerated by re-running RASAero II on independently reconstructed input decks. They are therefore a fixed, version-locked commercial reference rather than a controlled

head-to-head rerun. Any paired comparison a user performs against the `apogee_thiswork_ft` column inherits this asymmetry and should be framed accordingly.

The MESOS 293K model value is the corpus’s largest single-flight error.. The higher-Mach of the corpus’s two two-stage vehicles (flight 25, MESOS 293K, the highest-Mach flight at Mach 4.33; the other two-stage flight is the AeroPac 104K Two-Stage at Mach 3.04) is predicted by the archived code release at 273,056 ft, a signed error of -6.96% —the largest single-flight error in the corpus. The value is reproducible (confirmed by running the MESOS flight in isolation) and is the value carried in the published CSV. An earlier deposited version carried a different, erroneous value for this single flight; it has no defensible derivation and is superseded by the version of record. The -6.96% value remains within the $\pm 10\%$ admission band, so the headline “25/25 within $\pm 10\%$ ” count is unaffected.

Two base-drag constants make the companion-paper headline partly in-sample.. Two base-drag scale constants used to generate the `apogee_thiswork_ft` column (`THICK_BL_K` and `SLENDER_BODY_K` in the drag calculator) are corpus-frozen: they were calibrated against the apogee residuals of a subset of these very flights (Raven, Rabia and its short-fin-can variant, Kinsel, and Torrent via the source diagnostic). The aggregate accuracy of the `this work` column on the full corpus is therefore *partly in-sample*, and any statistic computed over all 25 flights should be read with that disclosure in mind. We do not re-argue the generalization question here; the companion research article [4] addresses it directly via a decontaminated prospective holdout (every flight any constant touched moved into a development set, leaving a genuinely blind holdout). Users who wish to evaluate the method out-of-sample should likewise exclude the calibration flights listed above before scoring.

Ethics Statement

The authors have read and followed the ethical requirements for publication in *Data in Brief*. This work did not involve human subjects, animal experiments, or data collected from social media platforms. All data in the Rocket Flight Database are drawn from openly published sources or are produced by the authors’ own simulations: the measured flight apogees and the paired RASAero II [1] reference predictions are the publicly released RASAero II altitude-comparison values [5], and the exploratory high-Mach

ground truth is taken from published flight-test reports (cited per row in the dataset). No third-party copyrighted material is redistributed beyond the fair, factual description of published numerical results needed to document the corpus; the externally sourced columns are attributed to their originators and are not represented as this work’s own measurements.

CRedit Author Statement

Aidan Yu: Conceptualization; Data curation; Software; Validation; Writing – original draft; Writing – review & editing.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data Availability

The Rocket Flight Database is openly archived on Zenodo under DOI 10.5281/zenodo.19976138 (CC-BY-4.0) [9], with a continuously updated mirror at <https://github.com/AidanSYu/rocket-flight-database>. The version of record described here is the synchronized release deposited at the time of submission, whose `apogee_thiswork_ft` column matches the current archived model code and the values printed in this article; the cited DOI resolves to that

synchronized deposit. Twenty-four of the twenty-five `apogee_thiswork_ft` values in the headline corpus are unchanged from the earlier deposited snapshot; the single exception is the MESOS 293K two-stage closure, for which the current archived model code predicts 273,056 ft (-6.96%). The earlier value carried for this single flight was erroneous and is corrected in the version of record (confirmed by running the flight in isolation); at -6.96% it remains within the $\pm 10\%$ admission band so the 25/25 headline is unchanged. The historical snapshot remains available as a prior version of the same Zenodo record. Companion analysis scripts and the full reproduction recipe are described in the dataset’s `reproducibility.md`; the scientific validation result is reported in the companion research article [4].

Generative AI Disclosure

During the preparation of this work the author used generative AI tools to assist with language editing, formatting, and code review only. All claims, equations, and numerical values were authored and verified by the human author, who takes full responsibility for the content of the publication. No AI system is listed as an author or contributor.

References

- [1] C. E. Rogers, RASAero II: Rocket aerodynamics analysis and flight simulation software, *Advances in the Astronautical Sciences* 154, aAS 15-367 (2015).
- [2] S. Niskanen, Development of an open source model rocket simulation software, Master’s thesis, Helsinki University of Technology, Espoo, Finland, available at <https://openrocket.info/documentation.html> (2009).
- [3] G. H. Ceotto, et al., RocketPy: Six degree-of-freedom rocket trajectory simulator, *Journal of Aerospace Engineering* 34 (6) (2021). doi:10.1061/(ASCE)AS.1943-5525.0001331.
- [4] A. Yu, An open-source integrated supersonic aerodynamic simulator for rocketry: 25-flight validation and comparison with a commercial baseline, *Journal of Spacecraft and Rockets* Companion research article (under review) (2026).

- [5] Rogers Aeroscience, RASAero II comparisons with altitude data, <https://www.rasaero.com/comparisons-alt.htm>, accessed 2026-05-16 (n.d.).
- [6] R. H. Heitkötter, Flight investigation of the performance of a two-stage solid-propellant Nike-Deacon (DAN) meteorological sounding rocket, Tech. Rep. NACA TN 3739, NACA (1956).
URL <https://ntrs.nasa.gov/citations/19930084525>
- [7] J. Galloway, Howard L., R. A. Crough, Nike Apache performance handbook, Tech. Rep. X-721-66-568, NASA Goddard Space Flight Center (1966).
URL <https://ntrs.nasa.gov/citations/19670015760>
- [8] Bristol Aerospace Limited, Black brant VB AAF-VB-32 flight performance, churchill research range, Tech. Rep. DTIC AD0733141, Defense Technical Information Center, exploratory-set reference (not in headline corpus); author list to be confirmed. (1971).
- [9] A. Yu, Rocket flight database (2026). doi:10.5281/zenodo.19976138.
URL <https://doi.org/10.5281/zenodo.19976138>
- [10] T. J. Sooy, R. Z. Schmidt, Aerodynamic predictions, comparisons, and validations using Missile DATCOM (97) and Aeroprediction 98 (AP98), Journal of Spacecraft and Rockets 42 (2) (2005) 257–265. doi:10.2514/1.7814.
- [11] A. M. Quintart, M. A. Haw, F. Semeraro, arcjetCV: Open-Source software to analyze material ablation, Journal of Spacecraft and Rockets 62 (5) (2025) 1644–1653. doi:10.2514/1.A36132.
- [12] Low-cost roll control authorship to be confirmed at proof stage, Development and flight validation of a low-cost rocket roll control system, Journal of Spacecraft and Rockets (2025). doi:10.2514/1.A36408.
- [13] COESA, U.S. Standard Atmosphere, 1976, Tech. Rep. NOAA-S/T 76-1562, NOAA, NASA, U.S. Air Force (1976).
URL <https://ntrs.nasa.gov/citations/19770009539>

- [14] J. S. Barrowman, The practical calculation of the aerodynamic characteristics of slender finned vehicles, Master's thesis, The Catholic University of America, Washington, DC (1967).
- [15] F. Wilcoxon, Individual comparisons by ranking methods, Biometrics Bulletin 1 (6) (1945) 80–83. doi:10.2307/3001968.